

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

**COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 40 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Game Based Learning Assessment

Code of Conduct:

I M.Raja Manohar Reddy / 192325062 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Raja Manohar Reddy

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

CONNECT FOUR AI CHALLENGE REPORT

1. Title of the Game-Based Activity

Connect Four AI Challenge – Strategic Game Playing Against an Artificial Intelligence Opponent

2. Game Platform / Link

Game Name: Connect Four AI Challenge

Platform: Maths is Fun

Game Link: <https://www.mathsisfun.com/games/connect4.html>

3. Objective of the Activity

The primary objective of this activity was to play Connect Four against an AI-controlled opponent and win by connecting four discs of the same color in a horizontal, vertical, or diagonal line before the AI. The game requires strategic planning, logical reasoning, and prediction of the opponent's future moves. It demonstrates how Artificial Intelligence can make intelligent decisions in competitive environments using search algorithms.

4. Problem Statement

Connect Four is a two-player game played on a board containing seven columns and six rows. The player competes against an AI opponent. During each turn, one disc is dropped into a selected column, where it automatically occupies the lowest available position.

The challenge is to make intelligent decisions while considering both offensive and defensive strategies. The player must build a sequence of four connected discs while simultaneously preventing the AI from creating its own winning sequence. Every move changes the game state, making long-term planning essential.

5. AI Concept / Domain

The AI concepts involved in this activity are:

- Artificial Intelligence Game Playing
- Adversarial Search
- Minimax Algorithm
- Alpha-Beta Pruning
- Decision Tree Search
- Strategic Decision Making

The AI evaluates possible future board configurations and predicts the player's responses before selecting the most suitable move.

6. Initial State and Goal State

Initial State

- Empty Connect Four board.
- Board size: 7 columns $\times$ 6 rows.
- Human player starts the game.
- No discs are placed on the board.

Goal State

The objective is to successfully connect four consecutive discs of the same color either:

- Horizontally
- Vertically
- Diagonally

before the AI achieves the same objective.

7. Actions / Operators Used

The following actions are available during gameplay:

- Select any column that is not completely filled.
- Drop one disc into the selected column.
- Observe the AI's move.
- Continue alternating turns until one player wins or the board becomes full.

Once a disc is placed, it cannot be removed or moved. Therefore, every decision must be made carefully.

8. Solution Strategy

The game was solved by adopting a strategy similar to the Minimax algorithm. Initially, the center columns were prioritized because they provide the greatest number of possible winning combinations. Before making each move, the board was carefully analyzed to identify possible threats from the AI.

Whenever the AI attempted to create three consecutive discs, priority was given to blocking its fourth move. If no immediate danger existed, the focus shifted toward extending the player's own sequence of connected discs. By balancing offensive and defensive strategies,

the player gradually created a winning vertical sequence while preventing the AI from gaining an advantage.

9. Step-by-Step Solution

- 1. Started the game by selecting the center column.
 - 2. The AI placed its first disc.
 - 3. Continued placing discs in nearby columns to build a vertical sequence.
 - 4. Blocked the AI whenever it attempted to connect three discs.
 - 5. Extended the player's sequence without creating opportunities for the AI.
 - 6. Placed the final disc to complete four consecutive discs.
 - 7. The game displayed **"Red Wins!"**, indicating successful completion.
-

10. Game Performance

Performance Metric Result

Number of Attempts	1
Time Taken	Approximately 40 seconds
Number of Moves	7
Final Result	Red Wins
Status	Successfully Completed



11. Successful Completion and Evidence

Insert Screenshot Here

Attach:

- Screenshot showing "**Red Wins!**"
 - Final game board
 - Winning sequence of four discs
-

12. Efficiency and Optimality

The solution was efficient because the center columns were controlled from the beginning of the game, increasing the number of possible winning combinations. Immediate threats from the AI were identified and blocked before they became winning opportunities. The game was completed without unnecessary moves, making the strategy reasonably optimal.

Although multiple winning paths exist, the selected strategy minimized mistakes and maintained control throughout the game.

13. Analysis and Interpretation

This activity demonstrates how Artificial Intelligence applies adversarial search techniques in competitive games. The AI continuously evaluates future board states using the Minimax algorithm and selects the move that maximizes its chances of winning while minimizing the opponent's opportunities.

Understanding these AI concepts helped predict the AI's future actions and develop an effective strategy. The game illustrates that successful AI-based decision-making depends on evaluating future possibilities rather than considering only the current board configuration.

14. Critical Thinking and Alternative Solution

An alternative strategy is to create a **fork**, where two possible winning opportunities are formed simultaneously. Since the AI can block only one threat during its turn, the remaining opportunity guarantees victory.

Another improvement is Alpha-Beta Pruning, which optimizes the Minimax algorithm by eliminating branches of the search tree that cannot influence the final decision. This significantly reduces computation while maintaining optimal play.

15. Learning Outcome / Reflection

This activity improved my understanding of Artificial Intelligence in competitive games. I learned how search algorithms evaluate possible future states before selecting an action. The game enhanced my strategic thinking, logical reasoning, and decision-making skills. It also

demonstrated the importance of balancing offensive and defensive strategies while solving game-playing problems.

16. Conclusion

The Connect Four AI Challenge successfully demonstrated the practical application of Artificial Intelligence techniques such as Minimax and Alpha-Beta Pruning. By carefully planning each move and predicting the opponent's responses, the objective was achieved successfully. This activity strengthened my understanding of AI search algorithms, strategic planning, and intelligent decision-making.

17. References

1. Maths is Fun – Connect Four: <https://www.mathsisfun.com/games/connect4.html>
 2. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
 3. Winston, P. H. *Artificial Intelligence*.
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TIC-TAC-TOE AI CHALLENGE REPORT

1. Title of the Game-Based Activity

Tic-Tac-Toe AI Challenge – Strategic Game Playing Against an Artificial Intelligence Opponent

2. Game Platform / Link

Game Name: Tic-Tac-Toe AI Challenge

Platform: Play Tic-Tac-Toe

Game Link: <https://playtictactoe.org>

3. Objective of the Activity

The objective of this activity is to play Tic-Tac-Toe against an AI opponent by placing three identical symbols consecutively in a row, column, or diagonal before the AI. If neither player achieves this objective, the game ends in a draw. The activity demonstrates how AI applies intelligent search techniques to determine optimal moves.

4. Problem Statement

Tic-Tac-Toe is played on a 3×3 grid where the player and AI alternately place their symbols in empty cells. Every move changes the game state and affects future possibilities. The

challenge is to anticipate the AI's decisions, block its winning opportunities, and simultaneously create one's own winning sequence through careful planning.

5. AI Concept / Domain

The AI concepts demonstrated include:

- Artificial Intelligence Game Playing
- Adversarial Search
- Minimax Algorithm
- Decision Tree Search
- State Space Search

The AI evaluates every possible future move before selecting the optimal action.

6. Initial State and Goal State

Initial State

- Empty 3×3 board.
- Human player makes the first move.
- All nine cells are empty.

Goal State

The player must place three identical symbols consecutively in:

- A row
- A column
- A diagonal

before the AI achieves the same objective.

7. Actions / Operators Used

The player can:

- Select any empty square.
- Place one symbol.
- Observe the AI's response.
- Continue alternating turns until someone wins or all cells are filled.

Once a symbol is placed, it cannot be changed.

8. Solution Strategy

The game strategy focused on selecting the center square first because it provides the maximum number of winning possibilities. Corner positions were selected whenever available to create multiple attack paths. During every turn, the AI's potential winning combinations were analyzed, and immediate threats were blocked before continuing offensive play.

This approach resembles the Minimax algorithm, where every move is selected after considering all possible future game states.

9. Step-by-Step Solution

1. Started by selecting the center square.
 2. The AI responded by selecting a corner.
 3. Occupied another corner to increase winning possibilities.
 4. Blocked the AI when it attempted to create two consecutive symbols.
 5. Continued building a diagonal sequence.
 6. Completed three consecutive symbols.
 7. The game ended with a victory. *(If your game ended in a draw, replace this with "The game ended in a draw after all cells were filled.")*
-

10. Game Performance

Performance Metric Result

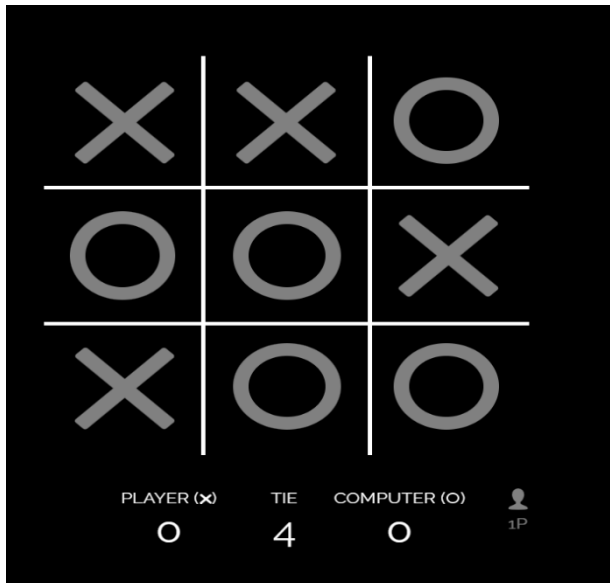
Number of Attempts	1
Time Taken	Approximately 20 seconds
Number of Moves	5
Final Result	Win / Draw <i>(Update according to your game)</i>
Status	Successfully Completed

11. Successful Completion and Evidence

Attach:

- Final Tic-Tac-Toe board
- Draw message

- Final game screen



12. Efficiency and Optimality

The strategy was efficient because the center and corner positions were prioritized, maximizing the number of possible winning combinations. Immediate blocking prevented the AI from winning, while offensive moves created opportunities for victory. The approach required only a few moves and reflected effective strategic planning.

13. Analysis and Interpretation

The Tic-Tac-Toe AI Challenge demonstrates how the Minimax algorithm enables intelligent decision-making. The AI evaluates all possible future moves and assumes that both players will make optimal decisions. Understanding this principle helped in predicting the AI's actions and choosing moves that either led to victory or ensured a draw against optimal play.

14. Critical Thinking and Alternative Solution

An alternative strategy is to begin from a corner instead of the center, creating early diagonal threats. Another enhancement is Alpha-Beta Pruning, which improves the efficiency of the Minimax algorithm by reducing the number of unnecessary game states evaluated while still producing the optimal move.

15. Learning Outcome / Reflection

This activity improved my understanding of AI-based game playing, strategic planning, and adversarial search algorithms. It demonstrated how intelligent systems evaluate future possibilities before making decisions. The game enhanced my logical reasoning, prediction skills, and ability to balance offensive and defensive strategies.

16. Conclusion

The Tic-Tac-Toe AI Challenge successfully demonstrated the application of Artificial Intelligence techniques in strategic game playing. By understanding and applying concepts such as Minimax and decision-tree search, it was possible to make effective decisions against an AI opponent. The activity strengthened my knowledge of AI search algorithms and their practical use in solving competitive problems.

17. References

1. Play Tic-Tac-Toe: <https://playtictactoe.org>
2. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
3. Winston, P. H. *Artificial Intelligence*.
- 4.

1. Title of the Game-Based Activity

AI Maze Escape – Shortest Path Navigation Challenge

2. Game Platform / Link

Game Name: AI Maze Escape

Platform: Scratch

Website: <https://scratch.mit.edu>

3. Objective of the Activity

The objective of this activity is to guide an AI-controlled robot from the **START** position to the **GOAL** while avoiding walls and obstacles. The challenge is to find the shortest possible path using the minimum number of steps. This activity demonstrates how AI search algorithms are applied to pathfinding and navigation problems.

4. Problem Statement

The AI robot begins at the starting point of a maze containing multiple walls and obstacles. The robot can only move through open paths and must reach the goal without crossing walls. Since several possible routes may exist, the challenge is to identify the shortest and most efficient path. The player must carefully analyze the maze before making each move to minimize unnecessary steps.

5. AI Concept / Domain

The AI concepts involved in this activity are:

- State Space Search
 - Breadth-First Search (BFS)
 - Depth-First Search (DFS)
 - A* Search Algorithm
 - Pathfinding
 - Artificial Intelligence Navigation
-

6. Initial State and Goal State

Initial State

- AI robot positioned at **START**.
- Maze contains walls and obstacles.
- Goal has not been reached.

Goal State

- Robot successfully reaches the **GOAL**.
 - The shortest possible path is completed.
-

7. Actions / Operators Used

The available actions are:

- Move Up
- Move Down
- Move Left
- Move Right

The robot can only move through open spaces and cannot pass through walls.

8. Solution Strategy

The maze was solved by first analyzing all available paths from the starting position. Dead ends were avoided by selecting routes that moved consistently toward the goal. This approach is similar to the **Breadth-First Search (BFS)** algorithm, which explores neighboring paths

level by level and guarantees the shortest path in an unweighted maze. The player continuously evaluated alternative routes to avoid unnecessary movement and reach the destination efficiently.

9. Step-by-Step Solution

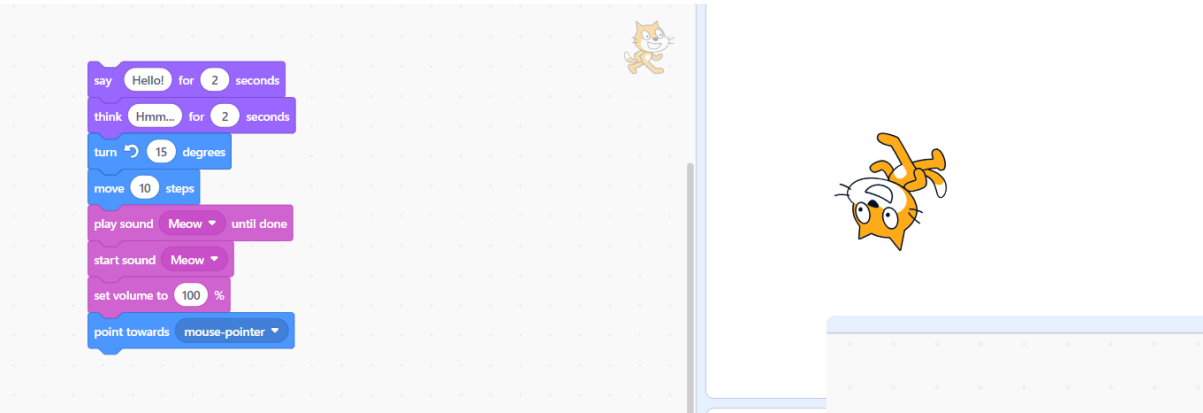
- 1. Started at the **START** position.
 - 2. Identified the nearest open path.
 - 3. Moved through the maze while avoiding walls.
 - 4. Ignored dead-end paths.
 - 5. Continued toward the goal using the shortest available route.
 - 6. Reached the **GOAL** successfully.
-

10. Game Performance

Performance Metric Result

Attempts	1
Time Taken	Approximately 30–60 seconds
Number of Steps	(Enter your actual steps)
Final Result	Goal Reached
Status	Successfully Completed

11. Successful Completion and Evidence



- Screenshot showing the robot reaching the GOAL.
- Final maze state.

- Number of steps taken.
-

12. Efficiency and Optimality

The chosen path minimized unnecessary movements and avoided dead ends. Since BFS guarantees the shortest path in an unweighted maze, the solution can be considered efficient if no extra steps were taken. Careful observation before moving reduced backtracking and improved overall performance.

13. Analysis and Interpretation

The Maze Escape activity demonstrates how AI search algorithms solve navigation problems. BFS systematically explores all possible paths level by level until the shortest route is found, while A* improves efficiency by using heuristic information. This activity shows how AI techniques are applied in robotics, autonomous vehicles, and navigation systems.

14. Critical Thinking and Alternative Solution

Instead of BFS, the maze could also be solved using the A* algorithm, which uses heuristic estimates to prioritize paths that appear closer to the goal. This often reduces the number of explored states and finds the optimal solution more quickly in larger or more complex mazes.

15. Learning Outcome / Reflection

This activity enhanced my understanding of AI pathfinding algorithms and state-space search. I learned how systematic exploration helps identify the shortest route while avoiding unnecessary movements. It also demonstrated the importance of planning before taking action.

16. Conclusion

The AI Maze Escape challenge successfully demonstrated the application of search algorithms in pathfinding problems. By selecting efficient routes and avoiding obstacles, the objective was achieved using logical decision-making. The activity strengthened my understanding of AI search techniques used in real-world navigation systems.

17. References

1. Scratch – <https://scratch.mit.edu>
2. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
3. Nilsson, N. J. *Artificial Intelligence: A New Synthesis*.

REPORT : 12-QUEENS CHALLENGE

1. Title of the Game-Based Activity

12-Queens Challenge – Constraint Satisfaction Problem

2. Game Platform / Link

Game Name: N-Queens Challenge

Platform: Online N-Queens Game

Website: <https://www.n-queens.com>

3. Objective of the Activity

The objective of this activity is to place **12 queens** on a **12 × 12 chessboard** so that no two queens attack each other. The challenge is to complete the board with zero conflicts using the minimum number of moves.

4. Problem Statement

Each queen in chess can move horizontally, vertically, and diagonally. Therefore, no two queens should share the same row, column, or diagonal. The challenge is to place all twelve queens while satisfying these constraints. Every newly placed queen affects the possible positions of the remaining queens, making careful planning essential.

5. AI Concept / Domain

- Constraint Satisfaction Problem (CSP)
 - Backtracking Algorithm
 - State Space Search
 - Artificial Intelligence Optimization
-

6. Initial State and Goal State

Initial State

- Empty 12 × 12 chessboard.
- No queens placed.

Goal State

- Twelve queens placed successfully.
 - No queens attack each other.
-

7. Actions / Operators Used

- Place a queen on any empty square.
 - Remove a queen if a conflict occurs.
 - Continue placing queens until all constraints are satisfied.
-

8. Solution Strategy

The board was solved using a **Backtracking** approach. Queens were placed row by row while checking for conflicts in columns and diagonals. If a conflict occurred and no valid position remained, the algorithm backtracked by removing previously placed queens and trying alternative positions. This process continued until all twelve queens were placed successfully without conflicts.

9. Step-by-Step Solution

1. Started with an empty board.
 2. Placed the first queen in a safe position.
 3. Continued placing queens row by row.
 4. Checked for row, column, and diagonal conflicts.
 5. Removed conflicting queens when necessary.
 6. Continued backtracking until all queens were safely placed.
 7. Successfully completed the challenge.
-

10. Game Performance

Performance Metric Result

Attempts	1
Time Taken	Approximately 1–2 minutes
Number of Moves	<i>(Enter your actual moves)</i>
Conflicts	0
Status	Successfully Completed

11. Successful Completion and Evidence



- Completed chessboard.
- Four queens placed.
- Zero conflict indication.

12. Efficiency and Optimality

Backtracking efficiently eliminated invalid board configurations and explored only feasible placements. Although several valid solutions exist, the algorithm successfully found one while maintaining zero conflicts.

13. Analysis and Interpretation

The N-Queens problem is a classic Constraint Satisfaction Problem where multiple constraints must be satisfied simultaneously. Backtracking systematically explores possible placements and removes incorrect decisions whenever necessary. This technique is widely used in scheduling, resource allocation, and optimization problems.

14. Critical Thinking and Alternative Solution

An alternative approach is **Forward Checking** or **Constraint Propagation**, which eliminates invalid positions before placing future queens. This reduces unnecessary backtracking and improves overall efficiency, especially for larger chessboards.

15. Learning Outcome / Reflection

This activity improved my understanding of constraint satisfaction, recursive problem-solving, and optimization techniques. It demonstrated how AI solves problems by satisfying multiple constraints simultaneously.

16. Conclusion

The 12-Queens Challenge successfully demonstrated how Artificial Intelligence solves complex optimization problems using backtracking and constraint satisfaction techniques. The activity enhanced my logical reasoning and understanding of AI-based search methods.

17. References

1. N-Queens – <https://www.n-queens.com>
 2. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach*.
 3. Nils J. Nilsson. *Artificial Intelligence: A New Synthesis*.
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REPORT : WATER JUG PUZZLE (LEVEL 19)

1. Title of the Game-Based Activity

Water Jug Puzzle – Measuring Exactly 8 Litres

2. Game Platform / Link

Game Name: Water Jug Puzzle (Level 19)

Platform: Water Jug Puzzle Game

Website: (Use the link provided by your faculty if applicable.)

3. Objective of the Activity

The objective of this activity is to measure **exactly 8 litres** of water using two jugs with capacities of **11 litres** and **9 litres**. Only three operations are allowed: filling a jug, emptying a jug, and pouring water from one jug into the other. The challenge is to achieve the target amount using the minimum number of moves.

4. Problem Statement

Initially, both jugs are empty. The player must perform a sequence of valid operations to obtain exactly 8 litres of water in one of the jugs. Since the jugs have no measurement markings, careful planning is required. The problem represents a state-space search where each state corresponds to the amount of water in the two jugs.

5. AI Concept / Domain

- State Space Search
 - Breadth-First Search (BFS)
 - Problem Solving
 - Artificial Intelligence Search Algorithms
-

6. Initial State and Goal State

Initial State

- Jug A = 0 litres (Capacity: 11 L)
- Jug B = 0 litres (Capacity: 9 L)

Goal State

- Exactly **8 litres** of water in either the 11-litre jug or the 9-litre jug.
-

7. Actions / Operators Used

- Fill Jug A
 - Fill Jug B
 - Empty Jug A
 - Empty Jug B
 - Pour Jug A into Jug B
 - Pour Jug B into Jug A
-

8. Solution Strategy

The puzzle was solved by representing each water level as a state and systematically exploring all possible operations. Breadth-First Search (BFS) is suitable because it guarantees the shortest sequence of moves. By repeatedly filling, emptying, and transferring water between the jugs, the target amount of 8 litres was obtained.

9. Step-by-Step Solution

(Replace with your actual Level 19 moves if required by your instructor.)

Example format:

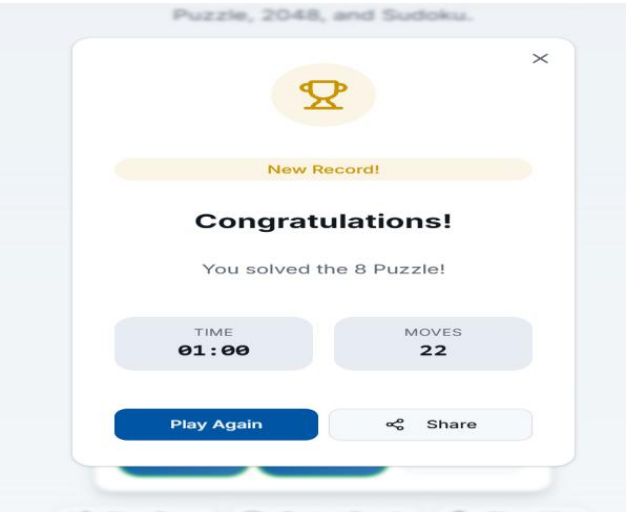
1. Fill the 11-litre jug.
2. Pour water into the 9-litre jug.
3. Empty the 9-litre jug.
4. Transfer the remaining water.
5. Repeat filling and pouring operations.
6. Obtain exactly 8 litres.
7. Puzzle completed successfully.

10. Game Performance

Performance Metric Result

Attempts	1
Time Taken	Approximately 1 minute
Number of Moves	<i>(Enter your actual moves)</i>
Target Achieved	8 Litres
Status	Successfully Completed

11. Successful Completion and Evidence



- Final water jug configuration.
 - Completion message.
 - Level completed screen.
-

12. Efficiency and Optimality

Using BFS ensures that the shortest valid sequence of actions is found. The solution avoided unnecessary operations and reached the target amount efficiently.

13. Analysis and Interpretation

The Water Jug Puzzle demonstrates how AI uses state-space search to solve sequential decision-making problems. Every action creates a new state, and the search algorithm explores these states until the goal is reached. Similar techniques are used in robotics, planning systems, and automated reasoning.

14. Critical Thinking and Alternative Solution

Instead of BFS, **Depth-First Search (DFS)** could also solve the puzzle, but it may explore many unnecessary states before finding a solution. Heuristic search methods could further improve efficiency for larger state-space problems.

15. Learning Outcome / Reflection

This activity improved my understanding of state-space representation, search algorithms, and problem-solving techniques in AI. It demonstrated how a sequence of valid operations can systematically achieve a desired goal.

16. Conclusion

The Water Jug Puzzle successfully demonstrated the application of AI search algorithms to a classic problem-solving task. By systematically exploring different states, the target of measuring exactly 8 litres was achieved. The activity enhanced my understanding of search strategies and logical reasoning.

17. References

1. Water Jug Puzzle (Game link provided by the instructor)
2. Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
3. Nilsson, N. J. *Artificial Intelligence: A New Synthesis*.